

Redes

Laboratorio 1: Esquemas de comunicación

The image shows a Wireshark network traffic capture window titled "intro-wireshark-trace1.pcap". The interface includes a toolbar at the top, a display filter bar, and a packet list table. The packet list shows several packets, with packet 280 selected. The packet details pane shows the structure of packet 280, which is an Ethernet II frame containing an Internet Protocol Version 4 packet and a Transmission Control Protocol (TCP) segment. The packet bytes pane shows the raw data of the selected packet in hexadecimal and ASCII.

No.	Time	Source	Destination	Protocol	Info	Protocol Length
275	13:22:51.767368	10.0.0.44	10.0.0.0	ICMPv6	Router Advertisement ...	174
276	13:22:52.504299	Sonos_25:3a:2a	Nearest-Customer-B...	STP	Conf. Root = 36864/0/...	60
277	13:22:53.528360	Sonos_25:3a:2a	Nearest-Customer-B...	STP	Conf. Root = 36864/0/...	60
278	13:22:53.734462	Maxlinear_80:00:00	Apple_98:d9:27	ARP	Who has 10.0.0.44? Te...	56
279	13:22:53.734575	Apple_98:d9:27	Maxlinear_80:00:00	ARP	10.0.0.44 is at 78:4f...	42
280	13:22:53.876033	10.0.0.44	128.119.245.12	TCP	53961 → 80 [SYN] Seq=...	78
281	13:22:53.877619	10.0.0.44	128.119.245.12	TCP	53962 → 80 [SYN] Seq=...	78
282	13:22:53.910856	128.119.245.12	10.0.0.44	TCP	80 → 53961 [SYN, ACK]...	76
283	13:22:53.910860	128.119.245.12	10.0.0.44	TCP	80 → 53962 [SYN, ACK]...	76

Frame 280: Packet, 78 bytes on wire (624 bits), 78 bytes captured (624 bits)

Ethernet II, Src: Apple_98:d9:27 (78:4f:43:98:d9:27), Dst: Maxlinear_80:00:00 (00:50:f1:80:00:00)

Internet Protocol Version 4, Src: 10.0.0.44, Dst: 128.119.245.12

Transmission Control Protocol, Src Port: 53961, Dst Port: 80, Seq: 0, Len: 0

```

0000  00 50 f1 80 00 00 78 4f 43 98 d9 27 08 00 45 00  .P....x0 C.....E.
0010  00 40 00 00 40 00 40 06 bb 08 0a 00 00 2c 80 77  .@...@...@.....,w
0020  f5 0c d2 c9 00 50 79 da f3 bd 00 00 00 00 b0 02  ....Py.....
0030  ff ff eb c1 00 00 02 04 05 b4 01 03 06 01 01  ....
0040  08 0a 0d b7 7d 21 00 00 00 00 04 02 00 00  ....}!...
  
```

intro-wireshark-trace1.pcap Packets: 651 Profile: Juan Cruz

```
juancruz@Mac:~/Universidad
~/Universidad
ifconfig
lo0: flags=8049<UP,LOOPBACK,RUNNING,MULTICAST> mtu 16384
    options=1203<RXCSUM, TXCSUM, TXSTATUS, SW_TIMESTAMP>
    inet 127.0.0.1 netmask 0xff000000
    inet6 ::1 prefixlen 128
    inet6 fe80::1%lo0 prefixlen 64 scopeid 0x1
    nd6 options=201<PERFORMNUD,DAD>
gif0: flags=8010<POINTOPOINT,MULTICAST> mtu 1280
stf0: flags=0<> mtu 1280
anpi0: flags=8863<UP,BROADCAST,SMART,RUNNING,SIMPLEX,MULTICAST> mtu 1500
    options=400<CHANNEL_IO>
    ether ee:3e:83:1c:97:c2
    media: none
    status: inactive
anpi1: flags=8863<UP,BROADCAST,SMART,RUNNING,SIMPLEX,MULTICAST> mtu 1500
    options=400<CHANNEL_IO>
    ether ee:3e:83:1c:97:c3
    media: none
    status: inactive
en3: flags=8863<UP,BROADCAST,SMART,RUNNING,SIMPLEX,MULTICAST> mtu 1500
    options=400<CHANNEL_IO>
    ether ee:3e:83:1c:97:a2
    nd6 options=201<PERFORMNUD,DAD>
    media: none
    status: inactive
en4: flags=8863<UP,BROADCAST,SMART,RUNNING,SIMPLEX,MULTICAST> mtu 1500
    options=400<CHANNEL_IO>
    ether ee:3e:83:1c:97:a3
    nd6 options=201<PERFORMNUD,DAD>
    media: none
    status: inactive
en1: flags=8963<UP,BROADCAST,SMART,RUNNING,PROMISC,SIMPLEX,MULTICAST> mtu 1500
    options=460<TS04,TS06,CHANNEL_IO>
    ether 36:02:a3:70:26:c0
    media: autoselect <full-duplex>
    status: inactive
en2: flags=8963<UP,BROADCAST,SMART,RUNNING,PROMISC,SIMPLEX,MULTICAST> mtu 1500
    options=460<TS04,TS06,CHANNEL_IO>
    ether 36:02:a3:70:26:c4
    media: autoselect <full-duplex>
    status: inactive
bridge0: flags=8863<UP,BROADCAST,SMART,RUNNING,SIMPLEX,MULTICAST> mtu 1500
    options=63<RXCSUM, TXCSUM, TS04, TS06>
    ether 36:02:a3:70:26:c0
    Configuration:
        id 0:0:0:0:0:0 priority 0 hellotime 0 fwddelay 0
        maxage 0 holdcnt 0 proto stp maxaddr 100 timeout 1200
        root id 0:0:0:0:0:0 priority 0 ifcost 0 port 0
        ipfilter disabled flags 0x0
    member: en1 flags=3<LEARNING,DISCOVER>
        ifmaxaddr 0 port 8 priority 0 path cost 0
    member: en2 flags=3<LEARNING,DISCOVER>
        ifmaxaddr 0 port 9 priority 0 path cost 0
    nd6 options=201<PERFORMNUD,DAD>
    media: <unknown type>
    status: inactive
utun0: flags=8051<UP,POINTOPOINT,RUNNING,MULTICAST> mtu 1380
    inet6 fe80::2cfd:539c:ac9e:a7cb%utun0 prefixlen 64 scopeid 0xf
```

Input

Output

Options

Capture to a permanent file

File: lab1_23110.pgcap

Browse...

Output format:

☒ pcapng

☐ pcap

Compression:

☒ None

☐ gzip

☐ LZ4

☒ Create a new file automatically...

☐ after

100000

packets

☐ after

5

megabytes

☐ after

1

seconds

☐ when time is a multiple of 1

hours

File infix pattern

☒ YYYYmmDDHHMMSS_NNNNN

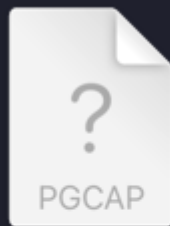
☐ NNNNN_YYYYmmDDHHMMSS

☒ Use a ring buffer with 10 files

Help

Close

OK



lab1_23110_2026
071422...01.pgcap

juancruz > Downloads

Capturing from Wi-Fi: en0

Apply a display filter ... <?/>

No.	Time	Source	Destination	Protocol	Info	Protocol Length
715	22:45:37.365153	192.168.31.152	142.251.34.138	UDP	53107 → 443 Len=32	74
716	22:45:37.924074	192.168.31.152	17.248.195.67	QUIC	Protected Payload (KP...	84
717	22:45:40.142054	192.168.31.134	255.255.255.255	UDP	56740 → 6667 Len=204	246
718	22:45:41.531239	192.168.31.152	224.0.0.251	MDNS	Standard query 0x0000...	87
719	22:45:41.531464	fe80::c58:3ac1:50c...	ff02::fb	MDNS	Standard query 0x0000...	107
720	22:45:41.531558	192.168.31.152	108.157.173.83	TLSv1...	Application Data	94
721	22:45:41.605371	108.157.173.83	192.168.31.152	TLSv1...	Application Data	90
722	22:45:41.605674	192.168.31.152	108.157.173.83	TCP	50845 → 443 [ACK] Seq...	66
723	22:45:41.819157	192.168.31.152	239.255.255.250	SSDP	M-SEARCH * HTTP/1.1	167

> Frame 1: Packet, 322 bytes on wire (2576 bits), 322 bytes captured (2576 bits) on interface en0, id 0
> Ethernet II, Src: XiaomiMobile_b6:e9:c8 (cc:d8:43:b6:e9:c8), Dst: da:13:a2:2a:75:cd (da:13:a2:2a:75:cd)
> Internet Protocol Version 4, Src: 142.251.34.138, Dst: 192.168.31.152
> User Datagram Protocol, Src Port: 443, Dst Port: 53107
> Data (280 bytes)

```
0000 da 13 a2 2a 75 cd cc d8 43 b6 e9 c8 08 00 45 00 ...*u... C...E-
0010 01 34 00 00 40 00 36 11 b1 f3 8e fb 22 8a c0 a8 ...4...@.6...-...
0020 1f 98 01 bb cf 73 01 20 3a 11 41 0d 60 2d 42 4c ...s...:A...-BL
0030 50 d4 6d 2f 39 ab 54 c4 cc 6c 10 41 40 08 99 9b P.m/9.T.L.AQ...
0040 ee 7a a4 46 68 91 20 36 45 2b 60 80 7c a3 bc 1c .z.Fh.6E+...|...
0050 d5 54 6a a2 f7 ec 8f 46 32 8f cf d4 0b b8 74 ef .Tj...F2...t.
0060 a1 13 25 62 a8 db 90 e1 4d 44 21 30 4e 3a 5a b7 ...%b...MD!0N:Z
0070 96 48 a9 64 0a f9 ed 2f a0 6f a1 92 19 98 25 2f .H.d.../o...%/
0080 7a 04 09 48 fa 1c da 93 86 f9 83 c0 3b 7c 4e 15 z.H...;|N
0090 e3 f7 5b 39 0a 44 53 b0 8e d8 53 27 4f c9 85 37 ..[9-DS..S'0..7
00a0 04 58 94 96 fd 45 ad 49 f5 e1 b9 cf 39 62 f3 bd .X...E.I...9b..
00b0 e4 4b f2 42 11 2a 38 23 3c 3a 3b 65 57 2f 6b 6b .K.B.*8# <;eW/kk
```

Wi-Fi: en0: <live capture in progress> Packets: 723 Profile: Juan Cruz

No.	Time	Source	Destination	Protocol	Info	Protocol Length
286	13:22:53.911521	10.0.0.44	128.119.245.12	HTTP	GET /wireshark-labs/I...	904
288	13:22:53.940406	128.119.245.12	10.0.0.44	HTTP	HTTP/1.1 200 OK (tex...	504
290	13:22:54.038247	10.0.0.44	128.119.245.12	HTTP	GET /favicon.ico HTTP...	861
292	13:22:54.062215	128.119.245.12	10.0.0.44	HTTP	HTTP/1.1 404 Not Foun...	551

Discusión:

En este laboratorio experimentamos las dos caras de la transmisión de datos. En la primera parte, comprobamos que los sistemas más simples (como el código Morse) generan menos errores al transmitirse. También vimos que incluir un conmutador es indispensable para ordenar los mensajes entre varios usuarios, aunque hace que la comunicación sea un poco más compleja.

En cuanto a la parte técnica, usar Wireshark fue un gran aprendizaje. Configurar el ring buffer me enseñó a hacer capturas de tráfico largas sin que la computadora se quede sin memoria. Además, al filtrar y analizar los paquetes HTTP, pude ver exactamente qué información se intercambia "por debajo" entre el navegador y el servidor al abrir una página web. En resumen, comprobé que este tipo de herramientas son esenciales para entender y arreglar problemas en cualquier red.